



**SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY**  
**COMPUTER ENGINEERING DEPARTMENT**  
**Proposal for the Final Year Project Batch 2020**



**1. Proposed Title**

**Augmented Reality-Based Computer Fundamentals Course**

**2. How would you categorize this project? [Tick only ONE]**

- a. An effort to facilitate an existing Research/ Industry Project. ✓
- b. An effort to facilitate developing a Research/ Industry Funded Proposal for possible Funding
- c. An effort to facilitate Master's/ PhD project objectives
- d. An exploratory study
- e. An experimental study for better lab utilization/ development
- f. A capstone project for students allowing them to apply their learning
- g. Other (please specify): \_\_\_\_\_

**3. What characteristics of the Complex Engineering Problem (CEP) are applicable to this project? [Tick one or more, if applicable]**

- a. Range of conflicting requirements: technical, engineering, or other issues. ✓
- b. Depth of analysis required: have no obvious solution and require abstract thinking.
- c. Depth of knowledge required: require research-based knowledge. ✓
- d. Familiarity of issues: Involve infrequently encountered issues ✓
- e. Extent of applicable codes: standards and codes devised by the professional bodies
- f. Extent of stakeholder involvement and level of conflicting requirements
- g. Consequences: Impact over the society. ✓
- h. Interdependence: problems due to dependability.

**4. Brief Outline**

In the 21st century, Augmented Reality (AR) technology has become increasingly important, due to its ability to enhance and augment our real-world experiences. AR can make learning more interactive and immersive, which may help to increase student engagement and motivation. Studies have shown that AR can be an effective tool for improving learning outcomes in a variety of subjects, including computer science. AR can provide an alternative to traditional teaching methods, making it easier for students with disabilities to access and participate in course content.



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### **5. Objectives**

The objective of an AR-based computer fundamentals course project might be to assess the effectiveness of using AR as a teaching tool for computer concepts to show its hardware and its devices in AR. This could involve comparing the learning outcomes of students who use the AR content with those who do not, or examining the engagement and motivation levels of students who use the AR content compared to those who do not.

### **6. Scope**

The future scope of an AR-based computer fundamentals course project to visualize the pictures of the course in 3D at a university could include expanding the use of 3D AR visualizations to other computer concepts or courses, replicating the project at other universities or with different student populations, conducting longitudinal studies to track the impact of 3D AR visualizations on learning outcomes over time, comparing the effectiveness of 3D AR visualizations with other technologies, and developing new 3D AR visualizations based on the results of the initial project. These various areas of future research could help to build upon the initial project and provide a deeper understanding of the potential benefits and limitations of using 3D AR visualizations in education at the university level.

### **7. Methodology**

The project will involve designing and developing the AR content, including the 3D visualizations of the course and the 3D models of computer components and devices. This may involve working with subject matter experts to ensure that the content is accurate and engaging. A key component of the project will be the creation of high-quality 3D models of computer hardware and devices. This will involve a 3D modeling software that is blender. Then, we will make an AR app through which student can view the computer related component in AR. The project will involve evaluating the effectiveness of the AR content in achieving the goals and objectives of the project. This may involve collecting data from students using surveys. The final step in the methodology will be to analyze and interpret the data collected during the evaluation phase, and to draw conclusions about the effectiveness of the AR content, including the 3D models of computer hardware and devices.



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**8. Expected Outcomes of the proposed project**

Also attach key milestone and deliverable chart here.

**Outcomes & Key Milestones:**

**→ Month 1:**

- Kick off meeting with the team
- Identify project scope and objectives
- Conduct market research

**→ Month 2:**

- Create project plan and timeline
- Design 3D models of computer hardware and devices
- Set up development environment and tools

**→ Month 3:**

- Implement basic AR features for 3D visualization of course pictures
- Test basic AR features

**→ Month 4:**

- Create 3D models of computer hardware and devices
- Integrate 3D models into AR content
- Test integration of 3D models

**→ Month 5:**

- Conduct pilot test with small group of students to identify any issues or areas for improvement

**→ Month 6:**

- Evaluate AR content using surveys.
- Analyze and interpret data collected during evaluation phase

**→ Month 7 & 8:**

- Address any issues or areas for improvement identified during evaluation
- Test updated AR content

**→ Month 9 & 10:**

- Launch AR-based computer fundamentals course
- Monitor performance and address any issues that arise
- Implement new features and functionality as needed



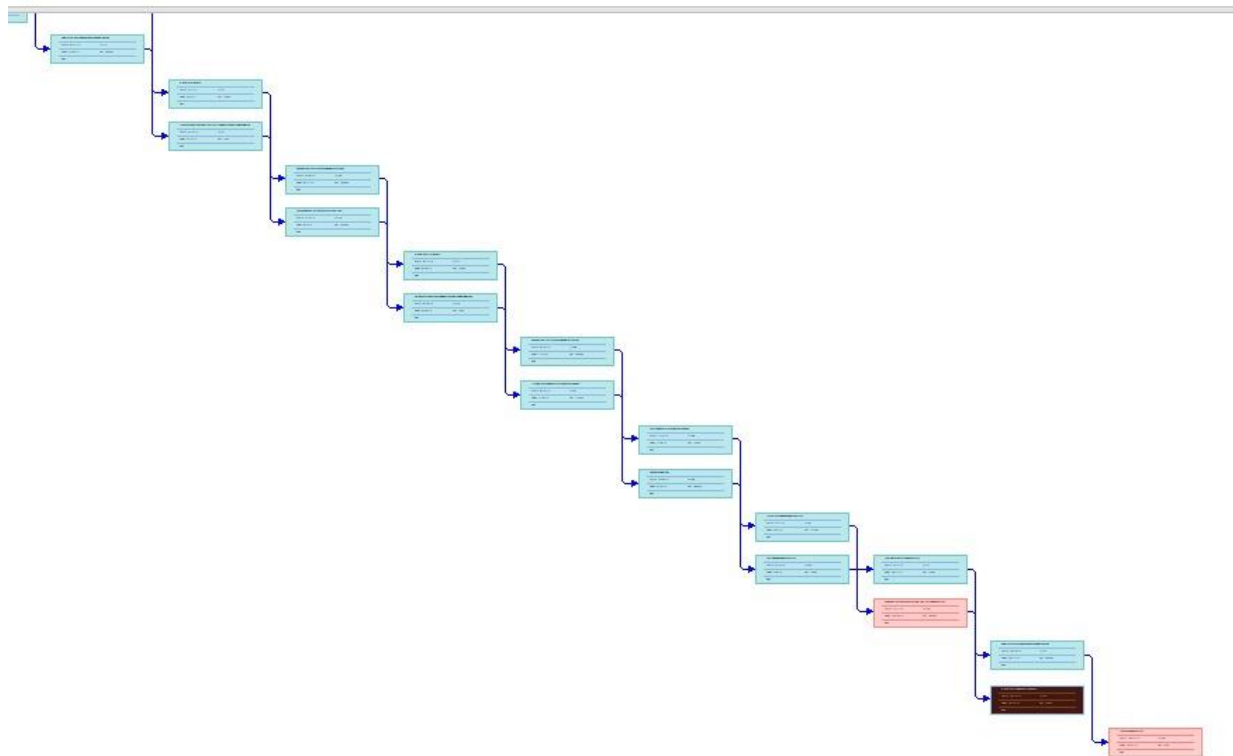
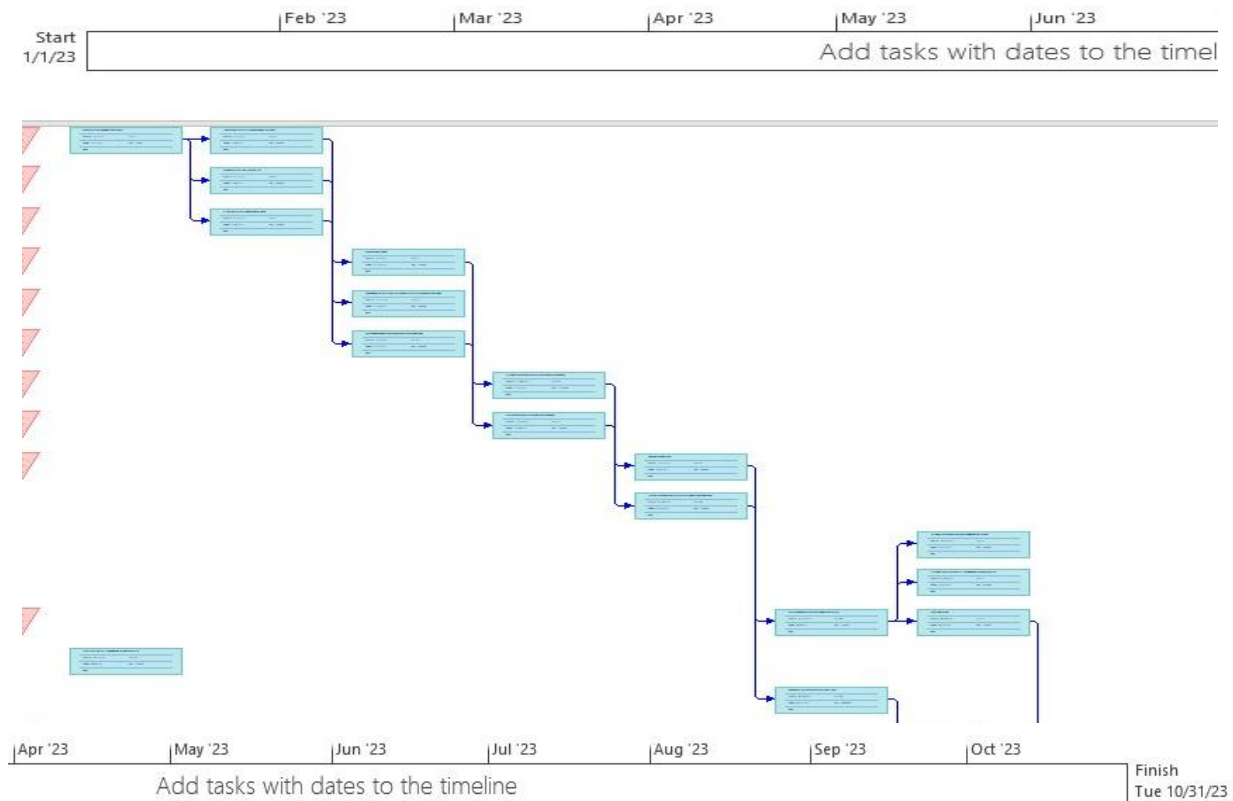
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#### Deliverable Chart:





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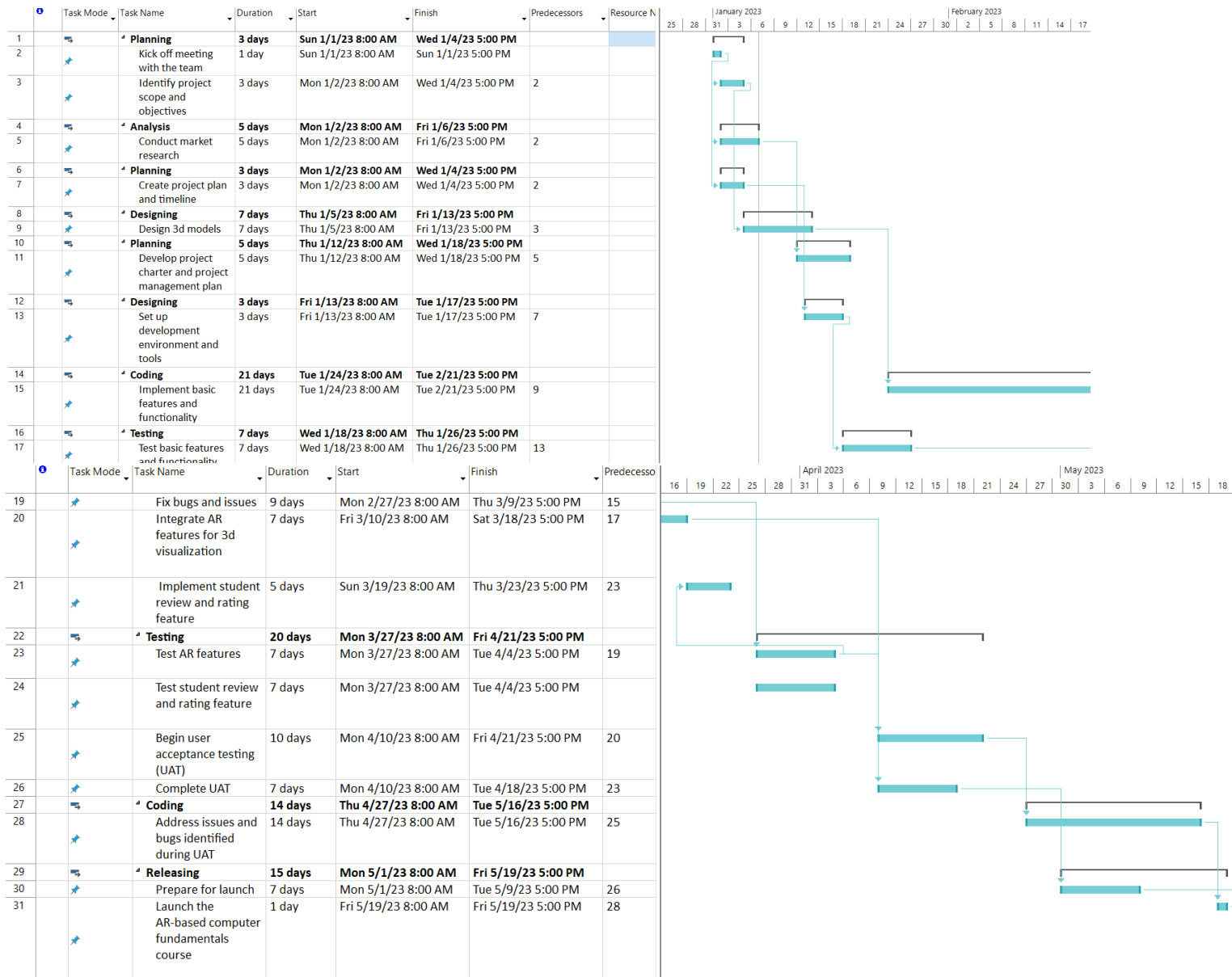


#### 9. Expected Budget

- Vuforia engine (99\$/month)
- Unity Assets (50\$-100\$)
- Some 3d models (50\$-100\$ as per use)

Total = 200\$-300\$ (45100PKR- 65000PKR approx.)

#### 10. Attach Gantt Chart for the Project Work

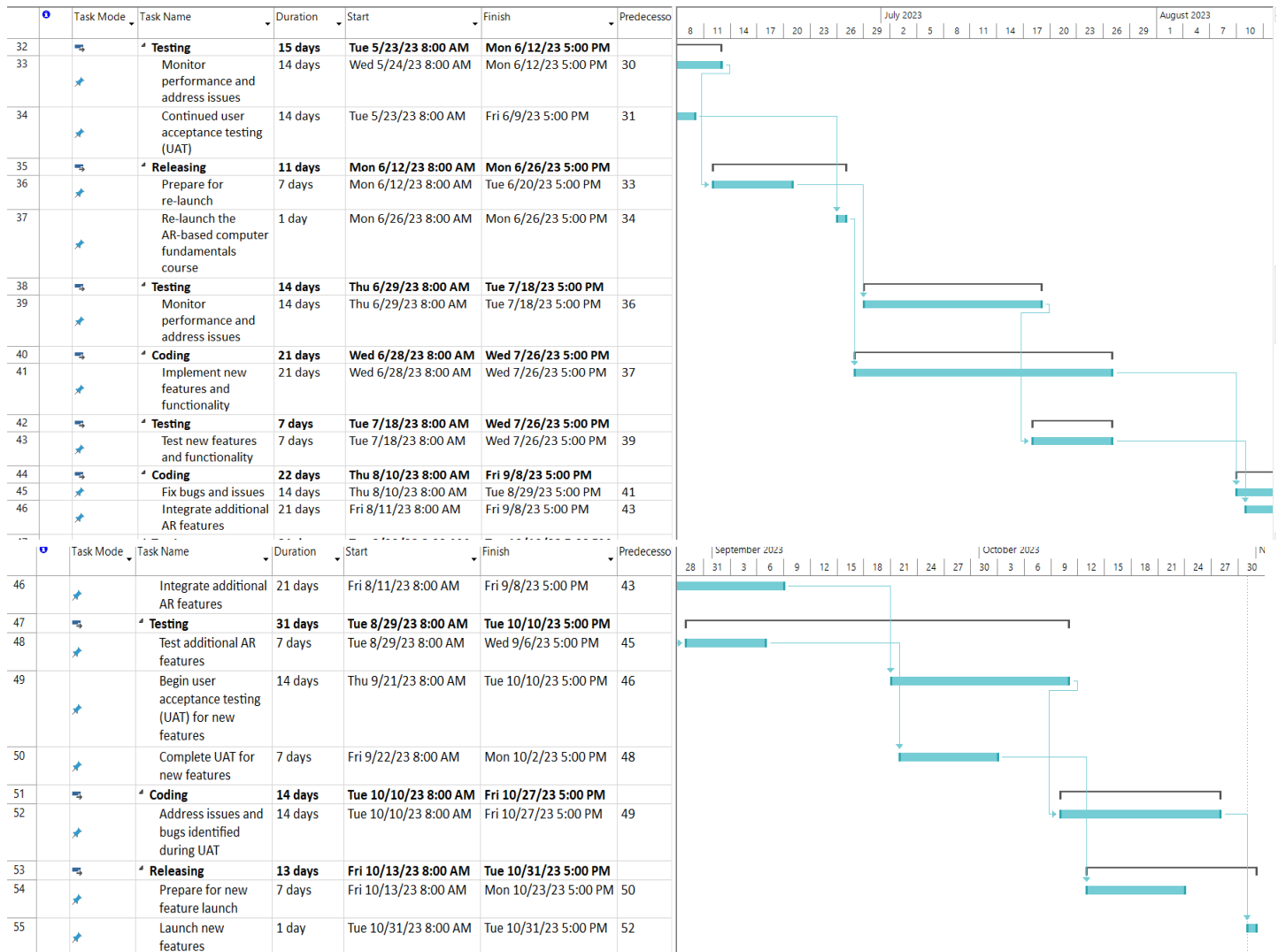




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#### 11. Hardware used

- Smartphone for visualizing AR 3d-model using AR app.

#### 12. Software tools used

- Blender for making 3d models
- Unity for implementing AR
- Vuforia engine for AR's database
- Visual studio

#### 13. Alignment of project with departmental/programme mission and SDGs.

##### → Alignment of project with Departmental Mission:

Augmented reality-based computer fundamentals course aligns well with the mission statement of a computer engineering department in several ways. Such as, it requires strong problem-solving and analysis skills, as the developer must consider how to design and implement to integrate AR technology effectively. Augmented reality-based computer fundamentals course can also lead to research, innovation, and entrepreneurship, For example, the developer may need to research and learn about the latest AR and working on this project can also help us to understand our moral and ethical responsibilities. Additionally, working on a team to build the AR based course can help us to develop our teamwork and leadership skills, aligning with the department's commitment to grooming us with these qualities.

##### → Alignment of project with SDG'S:

- **SDG 4: Quality Education: –**

By using AR technology to enhance the learning of computer science concepts, the project can contribute to the goal of providing inclusive and equitable quality education and promote lifelong learning opportunities for all.

- **SDG 9: Industry, Innovation, and Infrastructure: -**

The use of AR technology in the project can support the goal of building resilient infrastructure, promoting sustainable industrialization, and fostering innovation.

- **SDG 17: Partnerships for the Goals –**

The project may involve partnerships with educators, experts in computer and AR technology to achieve its goals and contribute to the SDGs.

#### **Reference of research paper:**

[https://www.researchgate.net/publication/281336331\\_A\\_Review\\_of\\_Research\\_on\\_Augmented\\_Reality\\_in\\_Education\\_Advantages\\_and\\_Applications](https://www.researchgate.net/publication/281336331_A_Review_of_Research_on_Augmented_Reality_in_Education_Advantages_and_Applications)



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Name of Supervisor: Mehjabeen Tahir Signature: \_\_\_\_\_

Name of Co- Supervisor (if any): \_\_\_\_\_ Signature: \_\_\_\_\_

Name of Industrial Advisor (if any): \_\_\_\_\_

**Recommendations and Approval:**

**Chairperson (Head of FYDP Review Committee) Sign with Date:**